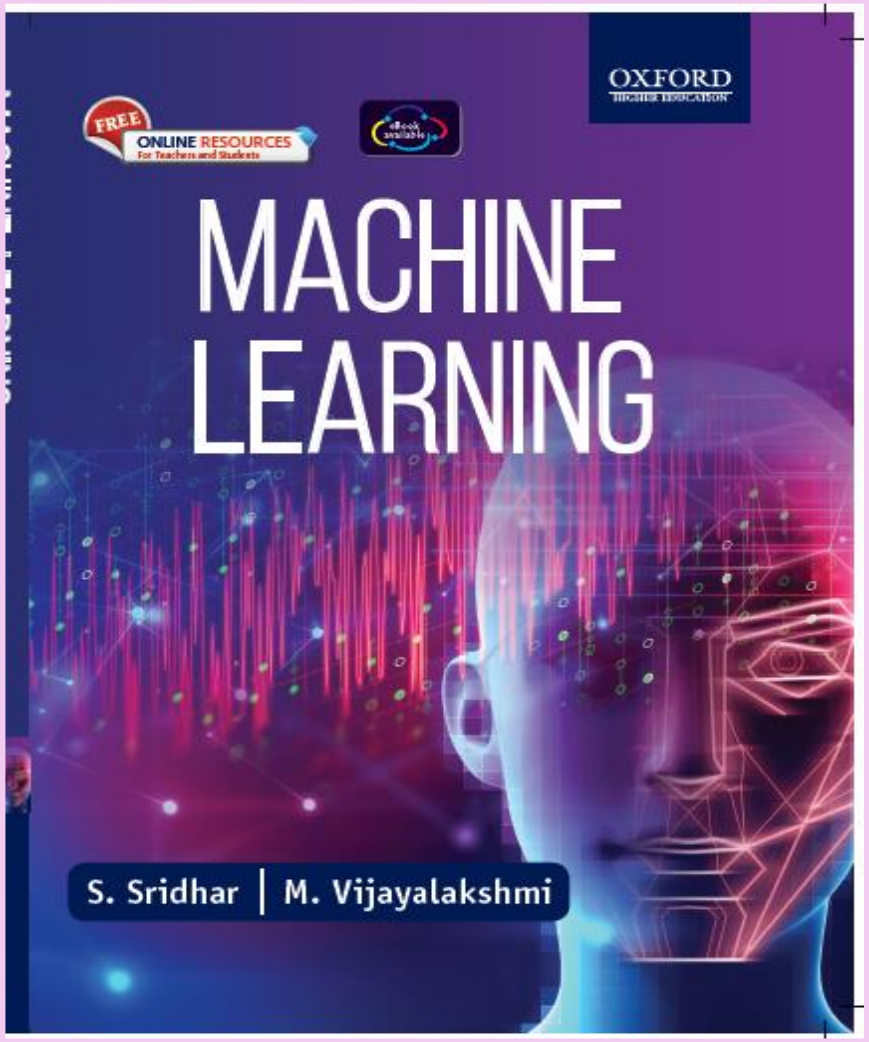


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Machine Learning

S. Sridhar and M. Vijayalakshmi

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Chapter 13

Clustering Algorithms

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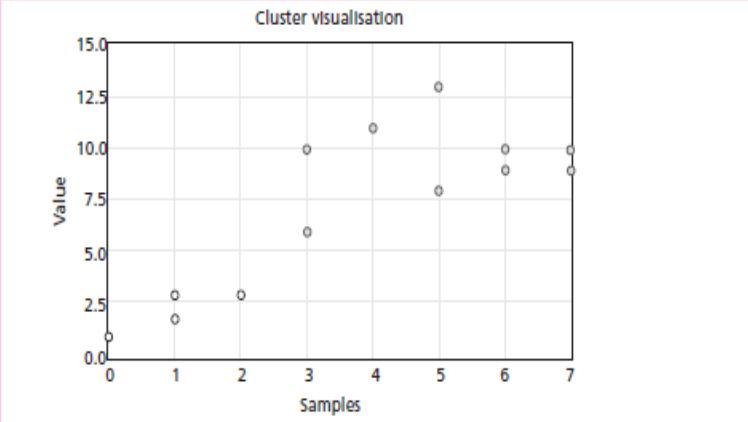
Clustering

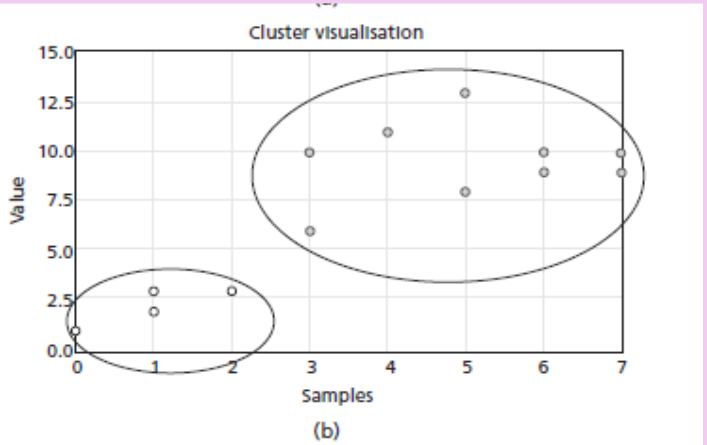
- CLUSTER ANALYSIS IS UNSUPERVISED LEARNING
- A TECHNIQUE OF PARTITIONING DATA INTO MEANINGFUL GROUP
- DONE USING TRIAL AND ERROR PROCESS

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Clustering

- CLUSTERING IS GROUPING PROCESS





(b)

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Difference between Clustering and Classification

S.No.	Clustering	Classification
1.	Unsupervised learning and cluster formation are done by trial and error as there is no supervisor	Supervised learning with the presence of a supervisor to provide training and testing data
2.	Unlabelled data	Labelled data
3.	No prior knowledge in clustering	Knowledge of the domain is must to label the samples of the dataset
4.	Cluster results are dynamic	Once a label is assigned, it does not change

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Applications of Clustering

Applications of Clustering

1. Grouping based on customer buying patterns
2. Profiling of customers based on lifestyle
3. In information retrieval applications (like retrieval of a document from a collection of documents)
4. Identifying the groups of genes that influence a disease
5. Identification of organs that are similar in physiology functions
6. Taxonomy of animals, plants in Biology
7. Clustering based on purchasing behaviour and demography
8. Document indexing
9. Data compression by grouping similar objects and finding duplicate objects

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Advantages and Disadvantages

Table 13.2: Advantages and Disadvantages of Clustering Algorithms

S.No.	Advantages	Disadvantages
1.	Cluster analysis algorithms can handle missing data and outliers.	Cluster analysis algorithms are sensitive to initialization and order of the input data.
2.	Can help classifiers in labelling the unlabelled data. Semi-supervised algorithms use cluster analysis algorithms to label the unlabelled data and then use classifiers to classify them.	Often, the number of clusters present in the data have to be specified by the user.

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Advantages and Disadvantages

S.No.	Advantages	Disadvantages
3.	It is easy to explain the cluster analysis algorithms and to implement them.	Scaling is a problem.
4.	Clustering is the oldest technique in statistics and it is easy to explain. It is also relatively easy to implement.	Designing a proximity measure for the given data is an issue.

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Properties of Distance Measures

The properties of the distance measures are:

1. D_{ij} is always positive or zero.
2. $D_{ij} = 0$, i.e., the distance between the object to itself is 0.
3. $D_{ij} = D_{ji}$. This property is called symmetry.
4. $D_{ij} \leq D_{ik} + D_{kj}$. This property is called triangular inequality.

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Euclidean Distance

The Euclidean distance between objects x_i and x_j with k features is given as follows:

$$\text{Distance}(x_i, x_j) = \sqrt{\sum_{k=1}^{n}(x_{ik} - x_{jk})^2} \quad (13.1)$$

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Manhattan Distance

$$\text{Distance}(x_i, x_j) = \sum_{k=1}^{n} \left| x_{ik} - x_{jk} \right|$$

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Chebyshev Distance

$$\text{Distance}(x_i, x_j) = \max_k \left| x_{ik} - x_{jk} \right|$$

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Example

Example 13.1: Suppose, if the coordinates of the objects are (0, 3) and (5, 8), then what is the Chebyshev distance?

Solution: The Euclidean distance using Eq. (13.1) is given as follows:

$$\text{Distance}(x_i, x_j) = \sqrt{(0-5)^2 + (3-8)^2}$$

$$= \sqrt{50} = 7.07$$

The Manhattan distance using Eq. (13.2) is given as follows:

$$\text{Distance}(x_i, x_j) = \left|(0-5) + (3-8)\right| = 10$$

The Chebyshev distance using Eq. (13.3) is given as follows:

$$\text{Max}\left\{\left|0-5\right|, \left|3-8\right|\right\} = \text{Max}\{5, 5\} = 5$$

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Binary Attributes

Table 13.3: Contingency Table

Attributes Matching	0	1
0	a	b
1	c	d

Simple Matching Coefficient (SMC) SMC is a simple distance measure and is defined as the ratio of number of matching attributes and the number of attributes. The formula is given as:

$$\frac{a+d}{a+b+c+d} \tag{13.5}$$

The values of a, b, c, and d can be observed from the Table 13.4.

Jaccard Coefficient Jaccard coefficient is another useful measure for and is given as follows:

$$J = \frac{d}{b+c+d} \tag{13.6}$$

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Example

Example 13.2: If the given vectors are $x = (1, 0, 0)$ and $y = (1, 1, 1)$ then find the SMC and Jaccard coefficient?

Solution: It can be seen from Table 13.2 that, $a = 0$, $b = 2$, $c = 0$ and $d = 1$.

The SMC using Eq. (13.5) is given as $\frac{a+d}{a+b+c+d} = 0+1/3 = 0.33$

Jaccard coefficient using Eq. (13.6) is given as $J = \frac{d}{b+c+d} = 1/3 = 0.33$

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Hamming Distance

Hamming Distance Hamming distance is another useful measure that can be used for knowing the sequence of characters or binary values. It indicates the number of positions at which the characters or binary bits are different.

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Categorical Variables

Categorical Variables

In many cases, categorical values are used. It is just a code or symbol to represent the values. For example, for the attribute Gender, a code 1 can be given to female and 0 can be given to male.

To calculate the distance between two objects represented by variables, we need to find only whether they are equal or not. This is given as:

$$\text{Distance}(x, y) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases} \quad (13.7)$$

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Ordinal Variables

Ordinal Variables

Ordinal variables are like categorical values but with an inherent order. For example, designation is an ordinal variable. If job designation is 1 or 2 or 3, it means code 1 is higher than 2 and code 2 is higher than 3. It is ranked as $1 > 2 > 3$.

$$\text{Distance}(X, Y) = \frac{|position(X) - position(Y)|}{n - 1}$$

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Cosing Similarity

$$sim(X,Y) = \frac{X \bullet Y}{\|X\| \times \|Y\|} = \frac{\sum_{i=1}^{n} X_i \times Y_i}{\sqrt{\sum_{i=1}^{n} X_i^2} \times \sqrt{\sum_{i=1}^{n} Y_i^2}}$$

Example 13.3: If the given vectors are $A = \{1, 1, 0\}$ and $B = \{0, 1, 1\}$, then what is the cosine similarity?

Solution: The dot product of the vector is $1 \times 0 + 1 \times 1 + 0 \times 1 = 1$. The norm of the vectors A and B is $\sqrt{2}$.

So, the cosine similarity using Eq. (13.9) is given as $\frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2} = 0.5$

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Hierarchical Clustering Algorithms

Algorithm 13.1: Agglomerative Clustering

1. Place each N sample or data instance into a separate cluster. So, initially N clusters are available.
2. Repeat the following steps until a single cluster is formed:
   (a) Determine two most similar clusters.
   (b) Merge the two clusters into a single cluster reducing the number of clusters as N–1.
3. Choose resultant clusters of step 2 as result.

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Example

Example 13.4: Consider the array of points as shown in the following Table. 13.4.

Table 13.4: Sample Data

Objects	X	Y
0	1	4
1	2	8
2	5	10
3	12	18
4	14	28

Apply single linkage algorithm.

Solution: The similarity table among the variables is computed as shown in Table 13.4. Euclidean distance is computed as shown in the following Table 13.5.

Table 13.5: Proximity Matrix

Objects	0	1	2	3	4
0	–	4.123	7.211	17.804	27.295
1		–	3.606	14.142	23.324
2			–	10.630	20.124
3				–	10.198
4					–

The minimum distance is 3.606. Therefore, the items 1 and 2 are clustered together. The resultant is shown in Table 13.6.

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Example

Table 13.6: After Iteration 1

Clusters	{1, 2}	0	3	4
{1, 2}	–	4.123	10.630	20.124
0		–	17.804	27.295
3			–	10.198
4				–

The distance between the group {1, 2} and items {0}, {3} and {4} is computed using the formula:

$$D_{SL}(C_i, C_j) = \text{minimum}_{a \in C_i, b \in C_j} d(a, b) \quad (13.10)$$

Here, D_{SL} is the single linkage distance, C_i, C_j are clusters and $d(a, b)$ is the distance between the elements a and b.

Thus, the distance between {1, 2} and {0} is:

Minimum {{1, 0}, {2, 0}} = Minimum {4.123, 7.211} = 4.123

The distance between {1, 2} and {3} is given as:

Minimum {{1, 3}, {2, 3}} = Minimum {14.412, 10.630} = 10.630

The distance between {1, 2} and {4} is given as:

Minimum {{1, 4}, {2, 4}} = Minimum {23.324, 20.124} = 20.124

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Example

This is shown in Table 13.6. The minimum distance of Table 13.6 is 4.123.

Therefore, {0, 1, 2} is clustered together. This result is shown in Table 13.7.

Table 13.7: After Iteration 2

Clusters	{0, 1, 2}	3	4
{0, 1, 2}	–	10.630	20.124
3		–	10.198
4			–

Thus, the distance between {0, 1, 2} and {3} using Eq. (13.10) is given as

Minimum {{0, 3}, {1, 3}, {2, 3}} = Minimum {17.804, 14.142, 10.630} = 10.630.

Thus, the distance between {0, 1, 2} and {4} is:

Minimum {{0, 4}, {1, 4}, {2, 4}} = Minimum {27.295, 23.324, 20.124} = 20.124.

This is shown in Table 13.7. The minimum is 10.198.

Therefore, the items {3, 4} are merged. And, the items {1, 2, 3} and {4, 5} are merged. The resultant is shown in Table 13.8.

Table 13.8: After Iteration 3

Clusters	{0, 1, 2}	{3, 4}
{0, 1, 2}	–	?
{3, 4}		–

The computation of ? in Table 13.8 is not needed as no other items are left. Therefore, the clusters {0, 1, 2} and {3, 4} are merged.

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Visualization

Dendrograms are used to plot the hierarchy of clusters. Dendrogram for the above clustering is shown in the following Figure 13.2.

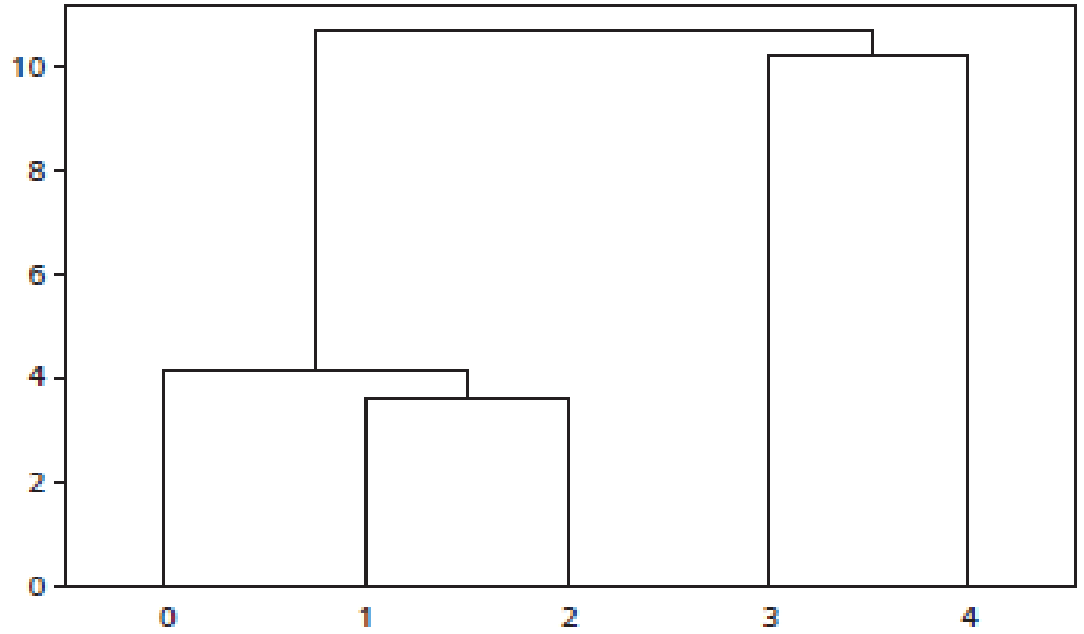


Figure 13.2: Dendrogram for Table 13.4

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Complete Linkage Algorithm

$$D_{CL}(C_i, C_j) = \text{maximum}_{a \in C_i,\, b \in C_j} d(a, b)$$

Dendrogram for the above clustering is shown in Figure 13.3.

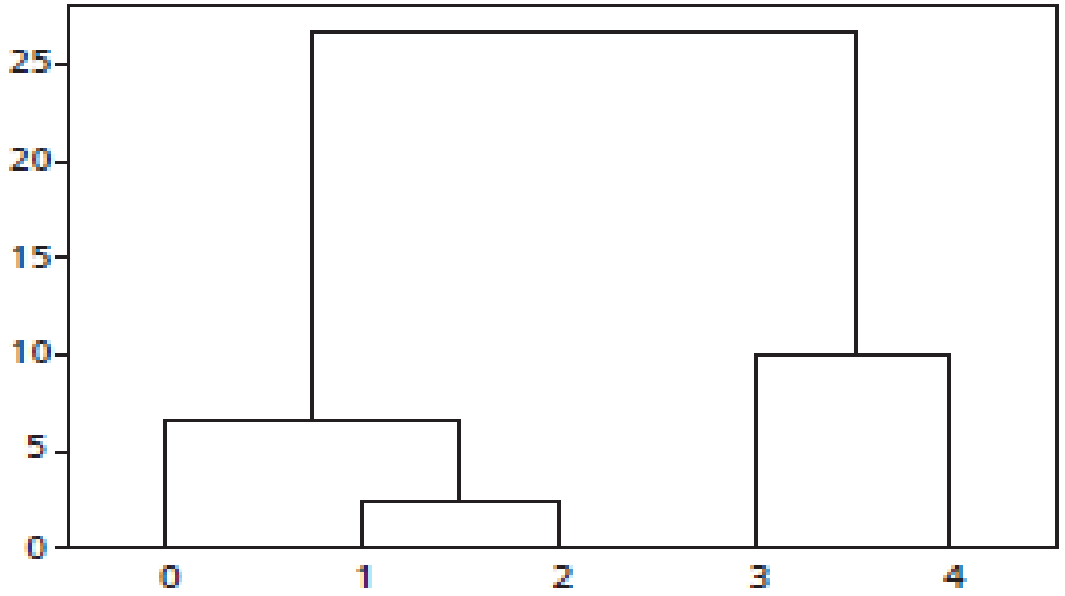


Figure 13.3: Dendrogram for Table 13.4

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Average Linkage

In case of an average linkage algorithm, the average distance of all pairs of points across the clusters is used to form clusters. The average value computed between clusters c_i, c_j is given as follows:

$$D_{AL}(C_i, C_j) = \frac{1}{m_i m_j} \sum_{a \in C_i, b \in C_j} d(a, b) \quad (13.12)$$

Here, m_i and m_j are sizes of the clusters.

The dendrogram for Table 13.4 is given in Figure 13.4.

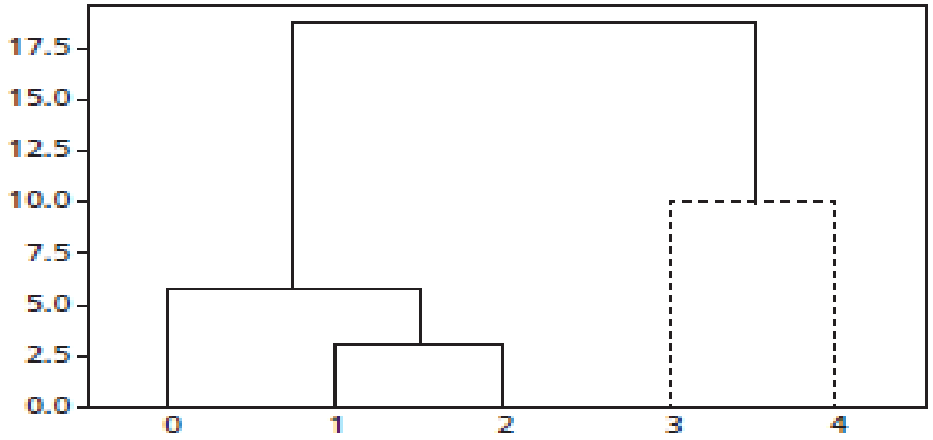


Figure 13.4: Dendrogram for Table 13.4

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Mean-Shift Algorithm

Algorithm 13.2: Mean-Shift Clustering

Step 1: Design a window.

Step 2: Place the window on a set of data points.

Step 3: Compute the mean for all the points that come under the window.

Step 4: Move the center of the window to the mean computed in step 3. Thus, the window moves towards the dense regions. The movement to the dense region is controlled by a mean shift vector. The mean shift vector is given as:

$$v_s = \frac{1}{K} \sum_{x_i \in S_k} (x_i - x) \tag{13.13}$$

Here, K is the number of points and S_k is the data points where the distance from data points x_i and centroid of the kernel x is within the radius of the sphere. Then, the centroid is updated as $x = x + v_s$.

Step 5: Repeat the steps 3–4 for convergence. Once convergence is achieved, no further points can be accommodated.

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Mean-Shift Algorithm

Advantages

1. No model assumptions
2. Suitable for all non-convex shapes
3. Only one parameter of the window, that is, bandwidth is required
4. Robust to noise
5. No issues of local minima or premature termination

Disadvantages

1. Selecting the bandwidth is a challenging task. If it is larger, then many clusters are missed. If it is small, then many points are missed and convergence occurs as the problem.
2. The number of clusters cannot be specified and user has no control over this parameter.

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K-means Algorithm

Algorithm 13.3: *k*-means Algorithm

Step 1: Determine the number of clusters before the algorithm is started. This is called k.

Step 2: Choose k instances randomly. These are initial cluster centers.

Step 3: Compute the mean of the initial clusters and assign the remaining sample to the closest cluster based on Euclidean distance or any other distance measure between the instances and the centroid of the clusters.

Step 4: Compute new centroid again considering the newly added samples.

Step 5: Perform the steps 3–4 till the algorithm becomes stable with no more changes in assignment of instances and clusters.

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Advantages and Disadvantages

Advantages

1. Simple
2. Easy to implement

Disadvantages

1. It is sensitive to initialization process as change of initial points leads to different clusters.
2. If the samples are large, then the algorithm takes a lot of time.

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Complexity of k-means Algorithm

Complexity

The complexity of k-means algorithm is dependent on the parameters like n, the number of samples, k, the number of clusters, $\Theta(nkld)$. I is the number of iterations and d is the number of attributes. The complexity of k-means algorithm is $O(n^2)$.

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Example

Example 13.5: Consider the following set of data given in Table 13.9. Cluster it using k-means algorithm with the initial value of objects 2 and 5 with the coordinate values (4, 6) and (12, 4) as initial seeds.

Table 13.9: Sample Data

Objects	*X*-coordinate	*Y*-coordinate
1	2	4
2	4	6
3	6	8
4	10	4
5	12	4

Solution: As per the problem, choose the objects 2 and 5 with the coordinate values. Hereafter, the objects' id is not important. The samples or data points (4, 6) and (12, 4) are started as two clusters as shown in Table 13.10.

Initially, centroid and data points are same as only one sample is involved.

Table 13.10: Initial Cluster Table

Cluster 1	Cluster 2
(4, 6)	(12, 4)
Centroid 1 (4, 6)	Centroid 2 (12, 4)

Iteration 1: Compare all the data points or samples with the centroid and assign to the nearest sample. Take the sample object 1 (2, 4) from Table 13.9 and compare with the centroid of

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Example

Dist (1, centroid 1) = $\sqrt{(2-4)^2 + (4-6)^2} = \sqrt{8}$

Dist (1, centroid 2) = $\sqrt{(2-12)^2 + (4-4)^2} = \sqrt{100} = 10$

Object 1 is closer to the centroid of cluster 1 and hence assign it to cluster 1. This is shown in Table 13.11. Object 2 is taken as centroid point.

For the object 3 (6, 8), the Euclidean distance between it and the centroid points is given as:

Dist (3, centroid 1) = $\sqrt{(6-4)^2 + (8-6)^2} = \sqrt{8}$

Dist (3, centroid 2) = $\sqrt{(6-12)^2 + (8-4)^2} = \sqrt{52}$

Object 3 is closer to the centroid of cluster 1 and hence remains in the same cluster 1.

Proceed with the next point object 4(10, 4) and again compare it with the centroids in Table 13.10.

Dist (4, centroid 1) = $\sqrt{(10-4)^2 + (4-6)^2} = \sqrt{40}$

Dist (4, centroid 2) = $\sqrt{(10-12)^2 + (4-4)^2} = \sqrt{4} = 2$

Object 4 is closer to the centroid of cluster 2 and hence assign it to the cluster table. Object 4 is in the same cluster. The final cluster table is shown in Table 13.11.

Obviously, Object 5 is in Cluster 3. Recompute the new centroids of cluster 1 and cluster 2. They are (4, 6) and (11, 4), respectively.

Table 13.11: Cluster Table After Iteration 1

Cluster 1	Cluster 2
(4, 6)	(10, 4)
(2, 4)	(12, 4)
(6, 8)	
Centroid 1 (4, 6)	Centroid 2 (11, 4)

The second iteration is started again with the Table 13.11.

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Example

Obviously, the point (4, 6) remains in cluster 1, as the distance of it with itself is 0. The remaining objects can be checked. Take the sample object 1 (2, 4) and compare with the centroid of the clusters in Table 13.12.

Dist (1, centroid 1) = $\sqrt{(2-4)^2 + (4-6)^2} = \sqrt{8}$

Dist (1, centroid 2) = $\sqrt{(2-11)^2 + (4-4)^2} = \sqrt{81} = 9$

Object 1 is closer to centroid of cluster 1 and hence remains in the same cluster. Take the sample object 3 (6, 8) and compare with the centroid values of clusters 1 (4, 6) and cluster 2 (11, 4) of the Table 13.12.

Dist (3, centroid 1) = $\sqrt{(6-4)^2 + (8-6)^2} = \sqrt{8}$

Dist (3, centroid 2) = $\sqrt{(6-11)^2 + (8-4)^2} = \sqrt{41}$

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Example

Object 3 is closer to centroid of cluster 1 and hence remains in the same cluster. Take the sample object 4 (10, 4) and compare with the centroid values of clusters 1 (4, 6) and cluster 2 (11, 4) of the Table 13.12:

Dist (4, centroid 1) = $\sqrt{(10-4)^2 + (4-6)^2} = \sqrt{40}$

Dist (3, centroid 2) = $\sqrt{(10-11)^2 + (4-4)^2} = \sqrt{1} = 1$

Object 3 is closer to centroid of cluster 2 and hence remains in the same cluster. Obviously, the sample (12, 4) is closer to its centroid as shown below:

Dist (5, centroid 1) = $\sqrt{(12-4)^2 + (4-6)^2} = \sqrt{68}$

Dist (5, centroid 2) = $\sqrt{(12-11)^2 + (4-4)^2} = \sqrt{1} = 1$. Therefore, it remains in the same cluster. Object 5 is taken as centroid point.

The final cluster Table 13.12 is given below:

Table 13.12: Cluster Table After Iteration 2

Cluster 1	Cluster 2
(4, 6)	(10, 4)
(2, 4)	(12, 4)
(6, 8)	
Centroid (4, 6)	Centroid (11, 4)

There is no change in the cluster Table 13.12. It is exactly the same; therefore, the *k*-means algorithm terminates with two clusters with data points as shown in the Table 13.12.

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Density-Based Clustering

1. Core point – A point is called a core point if it has more than specified number of points (m) within ε-neighbourhood.
2. Border point – A point is called a border point if it has fewer than 'm' points but is a neighbour of a core point.
3. Noise point – A point that is neither a core point nor border point.

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Measures

The following connectedness measures are used for this algorithm.

1. Direct density reachable – The point X is directly reachable from Y, if:
   (a) X is the ε-neighborhood of Y
   (b) Y is a core point
2. Densely reachable – The point X is densely reachable from Y, if there is a set of core points that leads from Y to X.
3. Density connected – X and Y are densely connected if Z is a core point and thus points X and Y are densely reachable from Z.

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DBSCAN Algorithm

Algorithm 13.4: DBSCAN

Step 1: Randomly select a point p. Compute distance between p and all other points.

Step 2: Find all points from p with respect to its neighbourhood and check whether it has minimum number of points m. If so, it is marked as a core point.

Step 3: If it is a core point, then a new cluster is formed, or existing cluster is enlarged.

Step 4: If it is a border point, then the algorithm moves to the next point and marks it as visited.

Step 5: If it is a noise point, they are removed.

Step 6: Merge the clusters if it is mergeable, $dist\ (c_i, c_j) < \varepsilon$.

Step 7: Repeat the process 3–6 till all points are processed.

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Advantages

Advantages

1. No need for specifying the number of clusters beforehand
2. The algorithm can detect clusters of any shapes
3. Robust to noise
4. Few parameters are needed

The complexity of this algorithm is $O(n \log n)$.

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Grid-Based Approaches

There are three important concepts that need to be mastered for understanding the grid-based schemes. They are:

1. Subspace clustering
2. Concept of dense cells
3. Monotonicity property

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Subspace Clustering

Grid-based algorithms are useful for clustering high-dimensional data, that is, data with many attributes. Some data like gene data may have millions of attributes. Every attribute is called a dimension. But all the attributes are not needed, as in many applications one may not require all the attributes. For example, an employee's address may not be required for profiling his diseases. Age may be required in that case. So, one can conclude that only a subset of features is required.

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Dense Cells

Algorithm 13.5: Dense Cells

Step 1: Define a set of grid points and assign the given data points on the grid.

Step 2: Determine the dense and sparse cells. If the number of points in a cell exceeds the threshold value τ, the cell is categorized as dense cell. Sparse cells are removed from the list.

Step 3: Merge the dense cells if they are adjacent.

Step 4: Form a list of grid cells for every subspace as output.

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CLIQUE

Algorithm 13.6: CLIQUE

Stage 1:

Step 1: Identify the dense cells.

Step 2: Merge dense cells c_1 and c_2 if they share the same interval.

Step 3: Generate Apriori rule to generate $(k + 1)^{th}$ cell for higher dimension. Then, check whether the number of points cross the threshold. This is repeated till there are no dense cells or new generation of dense cells.

Stage 2:

Step 1: Merging of dense cells into a cluster is carried out in each subspace using maximal regions to cover dense cells. The maximal region is an hyperrectangle where all cells fall into.

Step 2: Maximal region tries to cover all dense cells to form clusters.

In stage two, CLIQUE starts from dimension 2 and starts merging. This process is continued till the n-dimension.

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Advantages

Advantages of CLIQUE

1. Insensitive to input order of objects
2. No assumptions of underlying data distributions
3. Finds subspace of higher dimensions such that high-density clusters exist in those subspaces

Disadvantage

The disadvantage of CLIQUE is that tuning of grid parameters, such as grid size, and finding optimal threshold for finding whether the cell is dense or not is a challenge.

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Fuzzy Clustering

Algorithm 13.7: Fuzzy C-Means

Step 1: Choose the clusters, k, randomly.

Step 2: Assign weights w_{ij} of objects to clusters randomly.

Step 3: Compute the centroid:

$$c_j = \frac{\sum_{i=1}^{m} w_{ij}^p x_i}{\sum_{i=1}^{m} w_{ij}^p} \quad (13.15)$$

Step 4: The Sum of Squared Error (SSE) is computed as:

$$SSE = \sum_{j=1}^{k} \sum_{i=1}^{m} w_{ij}^p dist\,(x_i, c_j)^2 \quad (13.16)$$

Step 5: Minimize SSE to update the membership weights. Here, p is a fuzzifier whose value ranges from 1 to ∞. This parameter determines the influence of the weights. If p is 1, then fuzzy-c acts like k-means algorithm. A large weight results in a smaller value of the membership and hence more fuzziness. Typically, p value is 2.

Step 6: Repeat steps 3–5 till convergence is reached, which mean when there is no change in weights exceeding the threshold value.

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Shape of Data

Advantages and Disadvantages of FCM

The advantages and disadvantages of the FCM algorithm are given in Table 13.13.

Table 13.13: Advantages and Disadvantages of FCM Algorithm

S.No.	Advantages	Disadvantages
1.	Minimum intra class variance	The quality depends on the initial choice of weights
2.	Robust to noise	Local minima rather than global minimum

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EM Algorithm

Algorithm 13.8: Expectation-Maximization

Step 1: Select the parameters randomly.

Step 2: In expectation stage, for each point the condition probability is computed.

Step 3: In maximization stage, the new parameters are computed.

Step 4: Repeat the steps 2–3 till change is minimal within the threshold value or parameters do not change at all.

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Evaluation of Clustering

1. INTERNAL
2. EXTERNAL
3. RELATIVE

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Evaluation of Clustering

Cohesion (or compact) measures how close the samples are inside the cluster. This ensures that the clusters are homogeneous. Cohesion is measured as sum of squared errors between the samples and the centroid. The within cluster sum is given as:

$$\sum_{k=1}^{N} \sum_{x_i \in C} (x_i - m_j)^2 \qquad (13.17)$$

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Evaluation of Clustering

Separation indicates how well a sample differs from other clusters. This is measured as the weighted sum of the differences of the centroid of the dataset and the centroid of the generated clusters. This is given as:

$$\sum_{x_i \in C} C_i (x - x_i)^2 \qquad (13.18)$$

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DUNN Index

$$Index = \frac{\alpha \times separation}{\beta \times compactness}$$

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Silhouette Coefficient

Silhouette coefficient combines both cohesion and separation. The Silhouette coefficient measures the average distance between clusters. It is given as follows:

$$s_i = \frac{b_i - a_i}{\max\{b_i, a_i\}} \tag{13.20}$$

Here, a_i is the distance between the sample and centroid of the same cluster and b_i is the distance between the sample and the nearest centroid. The silhouette coefficients of the individual objects can be summed to get for the entire cluster as S, given as:

$$S = \frac{1}{N}\sum_{i=1}^{n} s_i \tag{13.21}$$

The value of the silhouette coefficient s_i is between –1 and +1. When it is closer to 1, the clusters are well formed. The value is zero when the data points are between two clusters and negative when the clusters are not formed correctly.

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Summary

1. Clustering is a technique of partitioning the objects with many attributes into meaningful disjoint subgroups.
2. Quantitative variables use distance measures, Euclidean distance, Manhattan distance and Chebyshev distance.
3. Binary attributes have only two values, 0 or 1. SMC and Jaccard distance are used to measure distance among objects with binary attributes.
4. To calculate the distance between two objects represented by nominal variables, techniques like simple matching, Jaccard method or Hamming distance are used.
5. Cosine similarity is a metric used to measure how similar the documents are irrespective of their size.

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Summary

6. Partitional methods use greedy procedures that are used iteratively to obtain a single level of partition.
7. Hierarchical methods produce a nested partition of objects with hierarchical relationships among objects. Often, the hierarchy relationships are shown in the form of a dendrogram.
8. *k*-means algorithm is an iterative partitional algorithm that uses the distance between the samples and the centroid of the clusters to find the closest clusters for cluster formation.
9. Density-based methods use the philosophy that atleast some minimum data points should be present within the acceptable radius for each point of the cluster.
10. Grid-based methods partition the data space into cells and use it for fitting the data.
11. Probability model-based methods use models to form clusters.
12. Purity is a measure of the extent to which a cluster consists of elements belonging to a single class.
13. The value of the silhouette coefficient S_i ranges between −1 and +1. When the formed clusters are compact, then the value of silhouette coefficient value approaches 1.

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